

Technical Craft & Code Interview: Preparation Guidelines

Candidate: Brent Rockwood

Interviewers: Chris Watts (CTO) & Jim

Duration: 60–75 minutes

Overview of the Session

Thank you for taking the time to speak during our initial screening call. For this next stage, we want to dive straight into the actual craft of your engineering work. We skip abstract whiteboard puzzles and generic testing frameworks in favor of a collaborative, peer-to-peer code review and architectural walkthrough.

We want you to guide us through production-grade code samples, infrastructure configurations, or architectural layouts that you have directly owned. Because our team operates on a "sole ownership" model where a principal engineer takes a project from start to finish, we want to look at how you structure code and conceptualize data flow at scale.

Key Themes & Code Deep Dives

During your walkthrough, please structure your code demonstration and repository tour around these core areas:

1. Node.js, TypeScript, Frontend and Backend Execution

We want to look at your practical application of JavaScript and TypeScript within modern UI and API ecosystems. Please prepare code snippets or architectures that demonstrate:

- Your comfort level structuring applications using modern libraries, specifically focusing on your experience with React, React Native, or Electron.
- How you leverage Node.js for backend systems such as GraphQL or REST APIs, Database communication, etc.

2. Python Engineering & Workflow Automation

Please show us examples of your Python work. We are particularly interested in seeing:

- Python experience includes building robust ingestion and orchestration pipelines for AI-assisted tooling at Rockwood Lab, developing sensor data workflows, and creating automation scripts for security platforms.
- Any other uses of Python you are proud of.

3. Candidate's Choice

Please present some examples of what you consider to be your best Senior or higher level

work in any language. Examples should demonstrate mastery of modern languages and concepts relevant to multi-user, data driven applications (API, Database, State, etc) or other tasks of similar complexity : 3D graphics programming, advanced comp-sci topics, etc.

Target Questions to Keep in Mind

As we step through your code, Jim and I will focus on your senior-level execution discipline and architectural foresight. Please come prepared to discuss:

- **System Design and Philosophy:** Be able to explain the principals that guide your system design choices and overall engineering philosophy.
- **Problem Solving Methodology:** How do you approach solving a novel problem?

A Note on Client NDAs: We respect intellectual property boundaries and regulatory constraints. To protect proprietary platforms, this session will be conducted entirely via live screen share. We do not require you to ship any files or repositories to us ahead of time, ensuring no permanent record of sensitive material is created.